



The ESO Astroquery module: a Jupyter Notebooks walkthrough

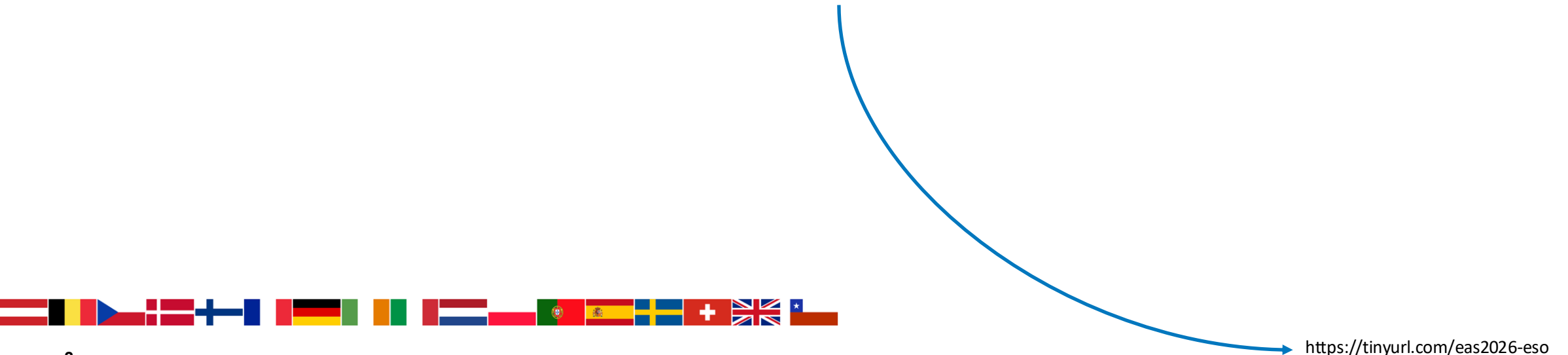
Ashley T. Barnes, Juan M. Carmona L. & Alberto Micol
DE & DS, ESO Back-end Operations Department



Hands-on session

Please download the slides from below

https://github.com/eso/astroquery_examples/blob/main/assets/eas2026-ucc-eso-astroquery.pdf





What is Astroquery?

Astroquery enables end-to-end analyses

 “Astroquery is a set of tools for querying astronomical web forms and databases.”

 **Documentation:** astroquery.readthedocs.io

 **Code and issue tracker:** <https://github.com/astropy/astroquery>

 **Paper:** Ginsburg, Sipőcz, Brasseur et al 2019.

Available Services

The following modules have been completed using a common API:

- ALMA Queries ([astroquery.alma](#))
- Atomic Line List ([astroquery.atomic](#))
- Besancon Queries ([astroquery.besancon](#))
- CADZ ([astroquery.cadc](#))
- CASDA Queries ([astroquery.casda](#))
- Cologne Database for Molecular Spectroscopy (CDMS) Queries ([astroquery.linealists.cdms](#))
- ESA EUCLID Archive ([astroquery.esa.euclid](#))
- ESA Herschel Science Archive ([astroquery.esa.hsa](#))
- ESA HST Archive ([astroquery.esa.hubble](#))
- ESA Integral Science Legacy Archive (ISLA) ([astroquery.esa.integral](#))
- ESA ISO Archive ([astroquery.esa.iso](#))
- ESA JWST Archive ([astroquery.esa.jwst](#))
- ESA XMM-Newton Archive ([astroquery.esa.xmm_newton](#))
- ESASky Queries ([astroquery.esasky](#))
- ESO Queries ([astroquery.eso](#))
- FIRST Queries ([astroquery.image_cutouts.first](#))
- Gaia TAP+ ([astroquery.gaia](#))
- GAMA Queries ([astroquery.gama](#))
- Gemini Queries ([astroquery.gemini](#))
- HEASARC Queries ([astroquery.heasarc](#))
- HiPS2fits Service ([astroquery.hips2fits](#))



- HITRAN Queries ([astroquery.hitran](#))
- IRSA Moving Object Search Tool ([astroquery.ipac.irsa.most](#))
- IRSA Dust Extinction Service Queries ([astroquery.ipac.irsa.irsa_dust](#))
- IRSA Image Server program interface (IBE) Queries ([astroquery.ipac.irsa.ibe](#))
- IRSA Queries ([astroquery.ipac.irsa](#))
- JPL Spectroscopy Queries ([astroquery.jplspec](#))
- MAGPIS Queries ([astroquery.magpis](#))
- MAST Queries ([astroquery.mast](#))
- CDS MOC Service ([astroquery.mocserver](#))
- Minor Planet Center Queries ([astroquery.mpc/astroquery.solarsystem.MPC](#))
- NASA ADS Queries ([astroquery.nasa_ads](#))
- NED Queries ([astroquery.ipac.ned](#))
- NIST Queries ([astroquery.nist](#))
- NVAS Queries ([astroquery.nvas](#))
- SIMBAD Queries ([astroquery.simbad](#))
- Skyview Queries ([astroquery.skyview](#))
- Splatalogue Queries ([astroquery.splatalogue](#))
- SVO Filter Profile Service Queries ([astroquery.svo_fps](#))
- UKIDSS Queries ([astroquery.ukidss](#))
- Vamdc Queries ([astroquery.vamdc](#))
- VizieR Queries ([astroquery.vizieR](#))
- VO Simple Cone Search ([astroquery.vo_conesearch](#))
- VSA Queries ([astroquery.vsa](#))
- xMatch Queries ([astroquery.xmatch](#))

The ESO module delivers complex datasets quickly

 “Simple python functions to script your access to the ESO archive via TAP.”

 ESO astroquery fork: <https://github.com/eso/astroquery>

 ESO example notebooks: https://github.com/eso/astroquery_examples

- *The ESO astroquery module allows users to search for raw and reduced data, retrieve metadata, access catalogues, and download data products. Proprietary access is also supported.*

WDB (“Old”)

- Limited query interfaces with predefined queries
- Users can only enter constraints on predefined queries
- No direct query language
- Users cannot modify underlying queries
- Catalogues not accessible
- Simplified access for specific, common queries

TAP (“New”)

- Full read-only access to ESO databases
- Users can write their own custom queries
- Uses ADQL 2.0 (SQL-based, VO standard)
- Users fully control query structure
- Catalogues accessible via dedicated interface
- Advanced, flexible querying for complex data retrieval

- *The **new** ESO module enables faster, larger and more complex dataset analyses by using the the Table Access Protocol (**TAP**), an International Virtual Observatory Alliance (**IVOA**) standard.*

Functionality

- List instruments and data releases.
- Instrument-specific and instrument-independent raw data search.
- Search across / by data releases.
- Automatic calibration selection (raw).
- Get extended FITS header information.
- Download datasets by their identifiers.
- Authenticated access to your assets.
- Run custom / complex ADQL queries.
- Retrieve atmospheric conditions data.

→ Automate all the above ←





**Where to get the new
ESO astroquery module and examples?**



Get the ESO module and examples from GitHub

Overview Repositories 6 Projects 1 Packages Teams People 32

ESO
20 followers International <http://www.eso.org> Unfollow

Popular repositories

- astroquery_examples** (Public)
Jupyter notebooks for querying, accessing, and analyzing data from the ESO Science Archive using astroquery.eso. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.
8 stars 3 forks
- astroquery** (Public)
Forked from [astro/astroquery](#)
Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8
Python 1 fork
- edps-doc** (Public)
TeX
- homebrew-pipelines** (Public)
A homebrew tap for ESO instrument pipelines
Ruby 3 stars 1 fork
- homebrew-test-pipelines** (Public)
A homebrew tap for ESO instrument pipelines (test version)
Ruby
- pyvo_examples** (Public)
Jupyter notebooks and scripts for querying, accessing, and analyzing data from the ESO Science Archive (mainly) using pyvo. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.
Jupyter Notebook

astroquery

- Fork of the official astroquery repository
- Latest ESO features not yet part of an official astroquery release



Get the ESO module and examples from GitHub

astroquery Public
forked from [astroquery/astroquery](#)

Edit Pins Watch 1 Fork 1 Star 0

develop-catalogues 7 Branches 0 Tags

Go to file Add file Code

This branch is 30 commits ahead of and 53 commits behind [astroquery/astroquery:main](#) Contribute Sync fork

ashleythomasbarnes	Remove temporary ROW_LIMIT handling	08882d8 · 2 months ago	10,831 Commits
· .github	Bump codecov/codecov-action from 5.5.2 to 6.0.0 in the a...	3 months ago	
· astroquery	Remove temporary ROW_LIMIT handling	2 months ago	
· docs	Merge pull request astroquery#3566 from bsipocz/DOC_fix_...	2 months ago	
· licenses	Moving license to root directory [skip ci]	9 years ago	
· .gitignore	made (hopefully) all changes from code review	4 years ago	
· .gitmodules	MAINT: removing AH	4 months ago	
· .mailmap	MAINT: update mailmap	9 months ago	
· .readthedocs.yaml	RTD: installing optional dependencies for RTD builds, too	4 months ago	
· CHANGES.rst	Update CHANGES.rst	2 months ago	
· CITATION	CITATION file should be in the source to be distributed to ...	7 years ago	

About

Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8

[astroquery.readthedocs.org/en/lat...](#)

Readme

BSD 3-Clause license

Contributing

Cite this repository

Activity

Custom properties

0 stars

1 watching

1 fork

Report repository

Releases

No releases published

[Create a new release](#)

astroquery

- **Important** for this example session we need the catalogue development version of astroquery

pip install

"git+https://github.com/eso/astroquery.git@develop-catalogues"



- **astroquery**
 - Fork of the official astroquery repository
 - Latest ESO features not yet part of an official astroquery release



Get the ESO module and examples from GitHub

Overview Repositories 6 Projects 1 Packages Teams People 32

ESO 20 followers International http://www.eso.org Unfollow

Popular repositories

astroquery_examples Public

Jupyter notebooks for querying, accessing, and analyzing data from the ESO Science Archive using astroquery.eso. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.

☆ 8 🍴 3

astroquery Public

Forked from [astropy/astroquery](#)

Functions and classes to access online data resources. Maintainers: @keflavich and @bsipocz and @ceb8

● Python 🍴 1

edps-doc Public

● TeX

homebrew-pipelines Public

A homebrew tap for ESO instrument pipelines

● Ruby ☆ 3 🍴 1

homebrew-test-pipelines Public

A homebrew tap for ESO instrument pipelines (test version)

● Ruby

pyvo_examples Public

Jupyter notebooks and scripts for querying, accessing, and analyzing data from the ESO Science Archive (mainly) using pyvo. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.

● Jupyter Notebook

View as: Public

You are viewing the README and pinned repositories as a public user.

People

View all

Top languages

● Python ● Ruby ● TeX

● Jupyter Notebook

Most used topics

astronomy eso eso-archive examples notebooks-jupyter

pyvo_examples

Jupyter notebooks with tutorials and real application examples making use of pyvo module.



Get the ESO module and examples from GitHub

astroquery_examples

- Jupyter notebooks with tutorials and real application examples

astroquery_examples Public

main 3 Branches 0 Tags

Go to file Add file Code

ashleythomasbarnes include more description of previews and updated .pngs c254bba · last month 77 Commits

File	Commit	Time
.binder	Update runtime.txt	4 months ago
assets	Upload eas2025 slides	last year
examples	include more description of previews and updated .pngs	last month
.gitignore	Moved around files...	10 months ago
LICENSE	Initial commit - README + LICENSE	last year
README.md	updated catalog files wording	9 months ago

README MIT license

ESO Science Archive - Jupyter Notebooks

⚠ Work in Progress — Updated: 22 August 2025

These notebooks are **under active development**. While we aim for accuracy, we **cannot guarantee** that everything in this repository currently works as intended.

Some functionality depends on features of `astroquery` that are **only available in the [ESO astroquery fork](#)**. Until these changes are merged into a stable release, please install `astroquery` directly from that branch to ensure compatibility.

Features, functions, and outputs may change without notice. Please report any issues via GitHub.

Introduction

This repository contains a suite of Jupyter notebooks designed to demonstrate how to explore, query, and retrieve data from the [ESO Science Archive](#) using Python. The primary interface used is the `astroquery.eso` module, which is part of the broader [astroquery](#) package developed under the Astropy ecosystem.

About

Jupyter notebooks for querying, accessing, and analyzing data from the ESO Science Archive using `astroquery.eso`. Includes examples for La Silla, Paranal (VLT), APEX, and ALMA datasets.

[archive.eso.org/cms.html](#)

[python](#) [examples](#) [astronomy](#)

[eso](#) [astroquery](#) [eso-archive](#)

[notebooks-jupyter](#)

Readme MIT license Activity Custom properties 8 stars 1 watching 3 forks Report repository

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Contributors 4

- ashleythomasbarnes** Ashley Bar...
- almicol** Alberto Micol
- gmella** Guillaume Mella





Get the ESO module and examples from GitHub

Name	Last commit message	Last commit date
..		
advanced	update dawn analysis with comments	9 months ago
case_studies	include more description of previews and updated .pngs	last month
simple	updated catalog files wording	9 months ago
README.md	Updated to highlights -> case_studies	10 months ago

Introduction to basic functionality

- Query the archive for raw* and reduced data
- Perform a simple cone search
- Obtain extended information on data products
- Download datasets from the archive

*Either instrument-specific or generic

Tutorials on basic functionality

- APEX products - reduced, raw, quick looks
- Searching via ADQL (SQL-like syntax)
- Downloading datasets by identifier
- Searching a source by name

Real life examples and use cases

- Find the closest observation to a given time
- Investigate ESO data using Aladin
- Download raw data for a reduced data product
- Examine Spectral Data as function of time



Get the ESO module and examples from GitHub

main ▾ astroquery_examples / examples / simple /

Go to file T Add file ▾

ashleythomasbarnes Add instrument-specific query example 9351e0d · 8 minutes ago History

Name	Last commit message	Last commit date
..		
00_quickstart.ipynb	Add instrument-specific query example	8 minutes ago
01_authentication.ipynb	checks for simple examples	last month
02_observations_query_reduced_by_position.ipynb	checks for simple examples	last month
03_observations_query_raw_by_position.ipynb	checks for simple examples	last month
04_observations_query_by_program_id.ipynb	checks for simple examples	last month
05_observations_query_observation_apex.ipynb	checks for simple examples	last month
06_observations_query_tap.ipynb	Add TAP cone, polygon and spatial examples	last month
07_observations_download_data.ipynb	checks for simple examples	last month
08_catalogs_query_with_constraints.ipynb	checks for simple examples	last month
09_catalogs_query_by_position.ipynb	checks for simple examples	last month
10_catalogs_query_tap.ipynb	updates to query_tap	last month
README.md	Replace introduction notebook with quickstart	16 minutes ago



Take a test drive

– Hands-on session –

Take a test drive on the cloud



Scroll down the github page and click here – **may take a little time to load up, so be patient...**

The image is a screenshot of a web browser displaying the README for the astroquery ESO Science Archive interface. The page has a light blue header with a 'README' tab and a 'MIT license' link. The main content is titled 'Introduction' and describes the repository's purpose: to demonstrate how to explore, query, and retrieve data from the ESO Science Archive using Python. It mentions the primary interface is the `astroquery.eso` module, part of the broader `astroquery` package. The text then lists the types of data products available, from raw data to analyzed spectra, and mentions the facilities used (La Silla, Paranal, VLT, VLTI, APEX, ALMA). It also states that the examples provide practical guidance for integrating archive queries into reproducible, scriptable workflows.

Below the introduction is a section titled 'Examples' with a sub-header 'All notebooks live under `examples/` and are grouped by purpose:'. A code block shows the directory structure of the `examples/` folder:

```
examples/
├── simple/           # 🌱 Minimal, single-feature recipes (copy/paste friendly)
├── advanced/        # 🚀 Scenario-driven workflows combining multiple functions
└── case_studies/    # 📁 Curated case studies for specific Phase 3/data collections
```

Below this, it says 'See detailed indexes inside each folder:' and lists three links: [examples/simple/README.md](#), [examples/advanced/README.md](#), and [examples/case_studies/README.md](#).

Next is a 'Quick Start' section with a 'launch binder' button. Below this, it says 'To get started, you can load and initialize the ESO archive interface directly in your notebook:' and shows a code block for loading the `astroquery.eso` module:

```
from astroquery.eso import Eso # Import the ESO module from astroquery
eso = Eso()                  # Create an instance of the ESO class
eso                           # Display the class instance (e.g. check login status, default
```

At the bottom, it says 'A quick-start notebook can be found in [examples/simple/00_quickstart.ipynb](#)'.

The screenshot shows a Jupyter Notebook environment for the ESO Science Archive. The interface includes a file explorer on the left, a main workspace with a notebook titled '00_quickstart.ipynb', and a terminal at the bottom. The notebook content covers the 'Quick Start' section, providing a workflow for using the archive, including authentication, data retrieval, and catalog searches. The code cells demonstrate how to import the necessary modules and check the version of astroquery.

Take a test drive on the cloud



The screenshot shows the GitHub interface for the repository `astroquery_examples`. The repository is public and has 1 branch and 0 tags. The file list includes `.binder`, `.devcontainer`, `.tools`, `assets`, `examples`, `.gitignore`, `AGENTS.md`, `LICENSE`, and `README.md`. The `README.md` file is selected, showing the ESO Science Archive - Jupyter Notebooks. The README includes a warning about work in progress and provides instructions for installing `astroquery` from the `develop-catalogues` branch of the ESO `astroquery` fork. The `Clone` dropdown menu is open, showing options for Local, Codespaces, and cloning via HTTPS, SSH, or GitHub CLI. An orange arrow points from the 'Clone' menu to a text box on the right.

You can also download locally and run the notebooks on your machine – **but make sure you have the correct version of astroquery installed**



Take a test drive on the cloud

astroquery_examples Public

main 3 Branches 0 Tags

Go to file

Add file

Code

Local

Codespaces

Codespaces

Your workspaces in the cloud

No codespaces

You don't have any codespaces with this repository checked out

Create codespace on main

Learn more about codespaces...

Codespace usage for this repository is paid for by ashleythomasbarnes.

ashleythomasbarnes include more description of previews and updates

- .binder Update runtime.txt
- assets Upload eas2025 sli
- examples include more descri
- .gitignore Moved around files.
- LICENSE Initial commit - REA
- README.md updated catalog file

README MIT license

ESO Science Archive - Jupyter Notebooks

For this example we will use github codespaces – which you can find by clicking on “code” and then “create codespace on main” - be patient this can take a little time to load...

Don't worry about this – its free if you have a github account!

Take a test drive on the cloud



The screenshot displays a web-based Jupyter Notebook environment. The top navigation bar shows the repository name 'astroquery_examples' and a 'glowing space adventure' theme. The left sidebar contains an 'EXPLORER' view with a file tree for 'ASTROQUERY_EXAMPLES', including folders like '.binder', 'assets', and 'examples', and files like '.gitignore', 'LICENSE', and 'README.md'. The main content area is titled 'ESO Science Archive - Jupyter Notebooks' and features a 'Work in Progress' warning dated 22 August 2025. The warning states that the notebooks are under active development and that some functionality depends on features of the 'astroquery' package that are only available in the 'ESO astroquery fork'. Below the warning is an 'Introduction' section explaining that the repository contains a suite of Jupyter notebooks for exploring, querying, and retrieving data from the ESO Science Archive using Python. The primary interface used is the 'astroquery.eso' module, part of the broader 'astroquery' package. The introduction also mentions that the notebooks walk through both introductory and advanced examples for a wide range of scientific use cases, from downloading raw and reduced data products to analyzing spectra and imaging datasets. At the bottom of the main content area, there is a terminal window showing the command prompt '@ashleythomasbarnes → /workspaces/astroquery_examples (main) \$'. The right sidebar contains a 'CHAT' section with a 'Build with Agent' button and a prompt to 'Describe what to build'. The bottom status bar shows the current branch as 'main' and the layout as 'U.S.'.

astroquery_examples [Codespaces: glowing space adventure]

EXPLORER

ASTROQUERY_EXAMPLES [C...]

- > .binder
- > assets
- > examples
- ◆ .gitignore
- 👤 LICENSE
- 📄 README.md

[Preview] README.md

ESO Science Archive - Jupyter Notebooks

⚠️ **Work in Progress — Updated: 22 August 2025**

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These notebooks walk through both introductory and advanced examples that are relevant for a wide range of scientific use cases — from downloading raw and reduced data products, to analyzing spectra and imaging datasets retrieved from ESO facilities at La Silla and Paranal (including the Very Large Telescope - VLT - and the Very Large Telescope Interferometer - VLTI), as well as from submillimeter and radio observatories -- APEX and ALMA.

Whether you're new to archival astronomy or already familiar with ESO's observing systems, these examples provide practical guidance for integrating archive queries into reproducible, scriptable workflows.

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

@ashleythomasbarnes → /workspaces/astroquery_examples (main) \$

CHAT

Build with Agent

AI responses may be inaccurate
[Generate Agent Instructions](#) to onboard AI onto your codebase.

Describe what to build

+ | 🔍 | Auto | ↕

Local | Default Approvals

Codespaces: glowing space adventure | main | 0 | 0 | 0

Sign In | Layout: U.S.

Thank you!

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@ESO Astronomy



@esoastronomy



@ESO



european-southern-observatory



@ESOobservatory

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